

PHOENIXGUARD V3 CONTINUOUS MARKET INTELLIGENCE

A source-grounded blueprint of how every proven closed candle becomes microstructure evidence, Pair DNA, a classified opportunity, an entry decision, and — only after independent validation — an external handoff.

Version: FINAL_LIVE / V3
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Streaming hardening
revision: 28 July 2026

EXECUTIVE POSITION

PhoenixGuard is a local-first chart intelligence and execution-control workstation. It must answer BUY_NOW, SELL_NOW or DO_NOT_ENTER honestly; silence and generic WAIT are not substitutes for a current explanation. A direction, confidence score, overlay, regression study, stream event or dashboard display is not trade authority.

This document describes the repository's active V3/FINAL_LIVE architecture and its declared boundaries. It is an engineering blueprint, not financial advice, a performance claim, or an instruction to place an order.

1. Executive blueprint

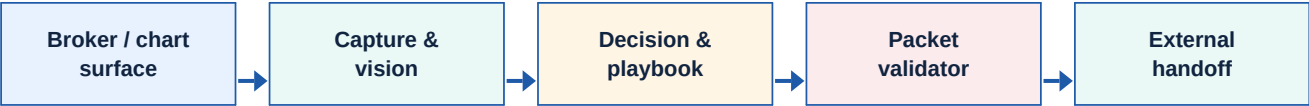
PhoenixGuard separates a high-risk workflow into independently accountable layers. It observes a locked broker/chart surface, reconstructs what is visible candle by candle, builds one explainable regression study from current and historical evidence, applies a rule-based playbook, exposes a plain-language operator story, and permits an external handoff only if a separate executable-packet contract succeeds.

CORE DOCTRINE

Observation ≠ study ≠ execution. The sole executable artefact is a fresh **PG_EXECUTION_PACKET_V3** carrying an explicit accepted **PG_ALLOWANCE_PACKAGE_V1**.

The layered design is intentional: screenshots can be stale; a visual model can be wrong; an overlay can be misaligned; historical similarity can be weak; and an apparent opportunity can be late. No single signal can jump directly to the execution edge.

The canonical authority chain



Stage	Question answered	Can it trade?
Observation	What is visibly on the locked chart?	No
Study	What conditions, paths and setups are plausible?	No
Permission	Does the current full context meet the playbook?	Not by itself
Packet validation	Is the exact permission current, internally consistent and safe to hand off?	Eligible only after pass
External boundary	Does the independent bridge accept the allowance package?	Separate downstream control

2. Runtime topology and source of truth

The canonical full-live launcher is **Backend/launch/launch_phoenixguard_live_ready.ps1**. It resolves the project-local live environment, prepares bounded runtime output, runs integrity preflight, starts the local stack, and verifies a single logical topology. The normal local API address is **127.0.0.1:8793**; the canonical session is **pocket-live-8788**.

Process / surface	Owns	Explicitly does not own
24/7 tracker orchestrator	Lifecycle, session creation, source locking and liveness	Trade permission in the launcher
FastAPI mobile API	Session state, capture services, public APIs and dashboard delivery	Browser-owned market truth
CPU stream observer	Continuous locked-surface sampling, bounded temporal evidence and event-gated keyframes	Running the heavy study for every sample or treating correlated frames as votes
Model Council + playbook	Evidence reconciliation, maturity and study/execution candidate output	Bypassing packet validation
shooter.py package reporter	Accepted allowance-package handshake	Clicking, calibration, amount edits or broker timing
MT4 / external bridge	Independent revalidation and optional file handoff	Weakening the V3 contract
Dashboard	Explainable rendering and commands	Truth, permission or closed-candle study progression

State and publication

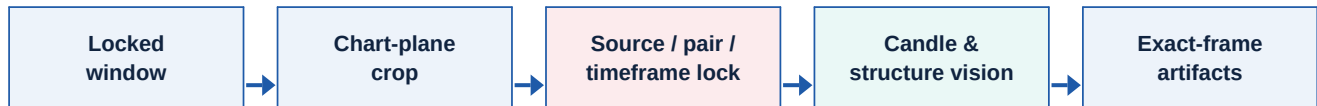
A capture is published through an atomic session commit. The exact frame’s source identity, chart/window artifacts, overlay geometry, study contributors, council result, public state, and closed-candle study transition are bound together. Stale writers are rejected so a slow result cannot overwrite a newer frame. Live session artifacts are ephemeral under runtime/live; durable study memory lives under data/mobile_api/window_tracker/market_study_v3.

FRONTEND CONTRACT

The dashboard renders three answers first: where the market came from, which direction was and is being studied, and whether to BUY_NOW, SELL_NOW or DO_NOT_ENTER. Details remain available below; the browser does not recalculate truth or permission.

3. How the chart is studied: capture, identity and reconstruction

Every live study begins with a real external broker/chart window. The tracker resolves the locked handle, captures the complete window, derives or applies a chart-plane crop, records window/chart hashes, confirms market and timeframe continuity, then fails closed if the source is missing, changed, stale or ambiguous.



Study output	How it is derived	Use downstream
Market identity	Broker surface + source lock + detected pair/timeframe	Rejects wrong-window / mixed-instrument truth
Candles	Visible candle geometry, current forming candle and latest confirmed closed candle	Exact micro-study, sequence identity and continuous history
Structure	Impulse, pullback, retest, continuation, swings, support/resistance, supply/demand	Playbook and price-location context
Chart transform	Exact window-space ↔ chart-space mapping	Keeps all overlays aligned with the frame
Quality / novelty	Parse confidence, ensemble disagreement, OOD / artifact checks	Caps confidence or makes entry DO_NOT_ENTER with a reason

Closed-candle identity bridge

Tracker track_id, generic id and rolling sequence indices are positional display identifiers, not durable candle identity. For screenshot-only rows without a stable source timestamp, persistence requires exact

PG_CLOSED_CANDLE_IDENTITY_STATE_V3 proof: a stable closed-candle event key plus a non-negative monotonic event sequence for the same pair/timeframe. The resolver promotes only its confirmed current/batch rows. A prior event becomes eligible for outcome study only when it is uniquely re-observed at the exact predecessor position on the current candle axis. Shifted, replayed or arbitrarily reacquired windows cannot manufacture that proof.

Overlay and positioning discipline

Raw detections are normalized to V3 object aliases, anchored to candle geometry, clipped to chart bounds, merged and collision-checked, then rendered only against their exact frame. Real source object keys remain distinct. Pixel bounds, touch points and anchor-wick points are normalized only with the exact captured image dimensions, after which raw pixel geometry is stripped. Anonymous objects receive observation-local **OBSERVATION_ONLY** identity. Graph edges are bounded study-only observations: never inferred causation, permission or an order. Entry, target and invalidation areas must be reconstructable from the anchor frame, preserve the correct price relationship, and be hidden rather than guessed when the global re-projection fit is not proven.

- **BUY_LIMIT_ZONE**: pullback opportunity below current price; **SELL_LIMIT_ZONE**: rally opportunity above current price.
- **BUY_STOP_ENTRY_ZONE** and **SELL_STOP_ENTRY_ZONE** are prospective confirmation boundaries, not interchangeable protective stops.
- **PROTECTIVE_STOP_ZONE** records plan invalidation with explicit broker-order side and thesis side.
- **NO_VALID_ZONE** is an explicit honest classification when evidence cannot support a causal, anchored area.

4. CPU-first continuous observation and event-gated study

Streaming in V3 means continuously observing the same verified chart surface while keeping expensive reconstruction and ensemble inference event-gated. The constrained CPU-only profile targets 0.5 samples per second and may be raised only after measurement; the supported range is 0.5 to 8 FPS. Those samples are temporal evidence, not independent opinions. The observer performs only bounded image normalization, stable hashing, compact grayscale differencing, identity checks and event classification; the existing heavy V3 study runs only when a material event or heartbeat keyframe is accepted.



Streaming stage	Bounded responsibility	Must never do
Capture producer	One process-global CPU stream acquires the locked HWND without activation; stamp monotonic and wall-clock times	Queue unbounded full-resolution frames, steal broker focus or silently switch windows
Light observer	Compute stable content digest, downsampled change, rest/motion evidence and identity generation	Run model council, count repeated samples as votes or create candle closes
Ring buffers	Retain only a tiny recent full-frame window plus compact grayscale evidence	Become a video archive or retain stale pair/geometry generations
Keyframe handoff	One pending slot plus one explicitly tracked in-flight keyframe; identity/geometry reset may replace pending work	Build latency, lose in-flight lineage or admit a second global producer
Heavy study worker	Consume one current keyframe, reconstruct the chart and publish through the atomic V3 commit	Overwrite newer state with a slow stale result
SSE workspace	Publish current stream health, three operator answers and exact-frame evidence	Infer truth or permission in the browser

GLOBAL SINGLE-FLIGHT / LATEST-FRAME-WINS

There is at most one process-global CPU producer. Its handoff has one pending slot and one separately tracked in-flight keyframe. While either is occupied, routine material and heartbeat admissions coalesce; sequence, metrics and drop counters still advance, so no hidden inference backlog forms.

Temporal evidence taxonomy

Observed class	Evidence recorded	Heavy-study behavior
DUPLICATE	Stable hash match or change below the duplicate floor; consecutive count and age advance	Discard; never a new vote, candle, trend or state transition
REST	Small but persistent bounded change consistent with a quiet/forming chart	Accumulate descriptive rest duration; study only if a material threshold or heartbeat is reached
MATERIAL_CHANGE	Change score, changed-area ratio or identity-safe motion exceeds the configured event threshold	Admit only when no pending/in-flight study exists; otherwise coalesce while recording metrics
HEARTBEAT	Maximum silence interval expires while identity remains valid	Admit only when idle; otherwise coalesce without erasing the supported decision
IDENTITY_RESET	Pair/selector, timeframe, HWND, focus/source lock, content size, DPI, ROI or geometry epoch changes	Break through the busy gate, increment generation, flush stale evidence/pending work and force rebind
CAPTURE_FAILURE	Window unavailable, minimized/occluded path invalid, focus ambiguous, capture exception or dimensions impossible	Fail closed, retain diagnostic reason, retry with bounded backoff; never reuse old pixels as current

Busy means coalesce, not stop observing. Every accepted CPU sample still advances lightweight frame sequence, stable hashes, motion/rest measurements, duration counters and identity/geometry comparison. Routine material changes and heartbeats cannot create a second heavy admission while pending or in-flight work exists. A genuine identity or geometry reset is evaluated first and may break through that gate so stale pair/coordinates are revoked immediately.

A quiet heartbeat is an observation about liveness, not a market verdict. Stream duplicates and degraded snapshot-watchdog duplicates preserve the current supported BUY_NOW, SELL_NOW or DO_NOT_ENTER explanation; they update health only and never overwrite it with generic WAIT. When entry gates are not satisfied, the public answer is the explicit **DO_NOT_ENTER** with a current reason—not an unexplained holding label.

Cross-process stream truth

The native tracker and HTTP API are separate processes. The tracker therefore publishes one atomic, replace-in-place **cpu_stream_v3.json** status sidecar at startup, approximately once per second, on degradation or recovery, and on stop. It contains bounded counters, ring and memory proof, monotonic lineage summaries and the current failure reason—never pixels or video. The public projection strips hashes, host paths, window identity, geometry internals and raw direction while explicitly denying broker-click authority. Compact session and operator reads merge this record without loading the large historical session document. Schema or session mismatch is rejected; unavailable or stopped telemetry fails closed. The forming-chart freshness window follows the slower of four target periods or three observed periods, with an eight-second floor and 45-second ceiling; it never refreshes closed-candle evidence or permission. Construction and capacity failures publish the same bounded record, so a producer that never starts still explains itself honestly.

Stable lineage and reset discipline

Each stream owns a random stream_id, a monotonic frame_seq and a stream_generation. Content digests use a stable cryptographic hash such as BLAKE2 or SHA-256 over canonical pixels and declared dimensions; Python process-local hash values are forbidden. Every accepted keyframe carries captured_epoch, monotonic capture time, HWND, source identity, pair, timeframe, window/content dimensions, DPI, chart ROI, geometry epoch, input_frame_hash and generation. This allows an exact frame to be traced through analysis, artifacts, operator state and packet diagnostics.

The stable visible pair/timeframe selector fingerprint contract hashes a thresholded bright-glyph mask from the broker's narrow selected-instrument control, excluding animated payout sparklines, clocks and candle pixels. The keyframe fingerprint must agree with its own full-window pixels. A change invalidates cached pair/timeframe reuse, forces a fresh selector scan and identity rebind, and keeps market identity pending until a normalized symbol is confirmed; the prior pair's candle field, Pair DNA and decision cannot leak into the new pair.

- A pair or timeframe change is a semantic reset even if the pixels look similar.
- An HWND or focus/source-lock change is a provenance reset even if the broker application is unchanged.
- A resize, DPI, chart-plane ROI or geometry transform change is a coordinate reset; old boxes and pending results become ineligible.
- A reset atomically clears recent evidence and the pending keyframe, increments generation and requires fresh identity proof before study resumes.
- A late heavy result is rejected if its generation, frame sequence or exact-frame lineage no longer matches current session truth.

Exact closed-candle causality inside a stream

Continuous pixels improve attention between accepted studies; they do not redefine what a completed candle is. Intrabar samples may update forming-candle movement, wick development, pressure, rest/motion duration and a watch thesis. Pair DNA, outcome maturation, motif promotion, regression history and any execution-eligible decision advance only from the existing authoritative closed-candle resolver. The current closed-candle event key, order domain and monotonic sequence must still prove exactly one new completed candle.

NO CORRELATED-FRAME VOTING

Ten near-identical frames from one forming candle remain one correlated observation stream. They cannot be counted as ten model votes, ten historical cases, ten supports or a confidence multiplier. Ensemble diversity must come from independently defined models or evidence lanes, not temporal duplication.

BOUND BEFORE CROP

The maximum pixel budget is checked against the complete captured window before any focus or chart crop. A small selected ROI cannot conceal an oversized full-window allocation from the CPU/RAM guard.

Capture trust order

Capture route	Acceptance rule	Failure behavior
Offscreen PrintWindow	High-rate route whenever the broker is not already foreground; exact HWND/title and broker pixels must validate	Never activates, raises or steals focus from the broker at stream cadence
Visible MSS / ImageGrab	Allowed only when the locked HWND already owns foreground before and after the exact-rectangle grab	Never activates the window; any ownership/title change falls back offscreen
Snapshot watchdog recovery	Separate low-frequency recovery may request bounded visible verification under explicit controls	Never becomes the CPU stream route or turns a duplicate into decision evidence
External frame feed	Study-only unless it independently proves the full local execution source contract	broker-click-safe remains false

The producer uses its own capture backend instance so thread-affine MSS state is not shared unsafely with the heavy worker. At stream cadence an offscreen chart is read only through validated PrintWindow; the stream never calls foreground activation. Desktop capture is allowed only when the locked broker already owns focus and still owns it after capture. On repeated failure the stream reports degraded/stopped health and uses bounded retry rather than spinning, switching screens or presenting stale pixels as live.

Three operator questions and the action boundary

Question shown first	Required plain-language answer	Evidence underneath
1. Where is the market from, and how did it behave?	Origin, major direction, swing/rest character and the relevant historical behavior	Closed-candle structure, behavioral sequence, Pair DNA and exact paths
2. What direction was studied, and what is being studied now?	Previous studied side, current major/inner direction, setup maturity and what would confirm or invalidate it	Regression, object relationships, current forming evidence and playbook
3. What is the best decision to do right now?	Enter a validated BUY/SELL, watch a pullback or rally, hold/protect, avoid chasing, or wait—with the decisive reason and change condition	Current entry gates, live stream read, entry areas, freshness, source/geometry identity, risk, council and packet state

The first viewport is a calm decision cockpit, not a field catalogue. Stream status is one compact line—observing, latest material event age, accepted/dropped keyframes and source health—with detailed telemetry collapsed. Labels-on may reveal every chart label, but the three answers remain readable and independent of overlay density.

BUY_NOW and SELL_NOW are operator-facing conclusions produced only when the playbook says ENTER_NOW and current evidence survives all entry gates. When entry is not currently authorized, Question 3 still gives the useful next action—watch the named pullback/rally area, hold or protect a confirmed trade, avoid chasing an expired move, or wait for one explicit condition. The answer must name whether the blocker is forming, insufficient, invalidated, late, stale, contradictory or runtime integrity. These words do not themselves click a broker. Only a fresh validated **PG_EXECUTION_PACKET_V3** with an accepted allowance may cross the external handoff boundary; direct local broker clicks remain disabled.

CPU and memory envelope

Resource	Target envelope on the CPU-only workstation	Control
Acquisition	0.5 FPS constrained-host baseline; one global stream across sessions	Configurable 0.5-8 FPS only after measurement; no second producer or overlapping capture tasks
Light processing	Small grayscale/downsample differencing and stable digest only	No model inference, OCR sweep or graph construction per raw sample
Full frames	Latest current frame plus only a very small bounded recent ring	Full-window pixel bound before focus crop; fixed item/byte cap and generation flush
Temporal data	A few seconds of compact grayscale summaries and counters	Fixed duration/count cap; no raw video persistence
Heavy study	Single-flight with one pending and one tracked in-flight slot	Busy material/heartbeat coalescing; reset breakthrough; stale-result rejection
Persistence	Significant keyframes, derived features, proof and closed-candle history; one replace-in-place stream-health sidecar	No continuous video archive; telemetry is one bounded schema-bound file

Failure recovery and certification

The stream is an optional V3 observation accelerator with a snapshot-compatible fallback. A producer crash, capture-path loss or observer exception cannot kill the API or manufacture a decision. The supervisor records the fault, clears ineligible pending work, backs off, reacquires the locked surface and starts a new generation. Restart recovery restores durable closed-candle study state, not stale in-memory pixel rings. Emergency stop and session stop terminate the producer and prevent further heavy-work wakeups.

Safe relaunch proves ownership before deleting disposable state. If managed Windows denies CIM/WMI command-line inspection, the canonical launcher falls back to psutil and selects a process only when both the exact resolved PhoenixGuard repository and a fixed runtime entrypoint token match. It excludes itself and its ancestors, validates the fresh V3 runtime lock while port 8793 is active, stops the attributed tracker/API, reporter, disk-guard and MT4-bridge trees, rescans for survivors, and independently proves port 8793 closed. Any uncertainty aborts before cleanup, so no unrelated Python process is terminated and no split stack is launched.

Shutdown is a service-wide lifecycle barrier. The shutdown event is set before workers are signalled. Stream ensure checks it before constructing a capture backend/observer and checks it again under the service lock immediately before registry insertion and thread.start(); a shutdown race closes the partly constructed backend and cannot leave a late producer running.

Certification layer	Required proof
Deterministic unit tests	Stable hashes and visible selector fingerprint; forced pair/timeframe rebind; duplicate/rest/material/heartbeat classification; identity/geometry reset; bounded rings
Concurrency/lifecycle tests	One global producer; one pending plus tracked in-flight; busy admission coalescing; reset breakthrough; stale-result rejection; shutdown-before-create/start race
Causality tests	Intrabar samples never advance Pair DNA or close history; exactly one identity-proven close advances once; repeated frames never increase vote/support counts
Geometry tests	Full-window pixel bound executes before focus crop; pair/HWND/focus/resize/DPI/ROI changes flush state; pixels, overlays and artifacts share dimensions
Performance tests	Measured light observation remains within the configured CPU-rate band and declared RAM caps while heavy study is single-flight and backlog remains at most one
API/browser tests	SSE and HTTP agree; three answers remain first; compact and cached operator reads receive fresh cross-process stream health; stale/tampered status fails closed; diagnostics remain collapsed
Live certification	Offscreen PrintWindow never activates broker at stream cadence; foreground capture identity, pair rebind, geometry reset, recovery and local-click disablement are observed

STREAMING AUTHORITY

Streaming can make PhoenixGuard more attentive and more timely. It cannot grant execution authority, bypass exact closed-candle causality, repair an invalid source lock, or convert repeated pixels into independent evidence.

5. Intelligence lanes: how a study is classified

The main image-analysis entry point is **Frontend/dashboard/main.py::run_inference**. It composes computer-vision detections, chart/sequence extraction, optional local ensemble output, memory retrieval, uncertainty models, reinforcement-learning context, 13 core skill gates plus support gates, and an ensemble decision. These models may remain internal contributors, but their old public forecast lanes are retired. Their outputs are inputs to study and council logic; they are not execution authority.

Lane	What it contributes	Authority
Computer vision	Detected patterns, candles, chart geometry, market state and reasoning trace	Observational
Scene / sequence models	Internal structure and sequence evidence from awake, versioned models	Internal contributor; retired as a public visual route
High-frequency / candle models	Near-term candle and movement context	Internal contributor; never an entry lane
Visual memory	Top-k analogous situations, similarity, ambiguity and transition tendencies	Advisory; disagreement can reduce confidence
A* scenario engine	Ranked continuation, pullback, reversal-attempt and fakeout paths	Decision-kernel evidence
Regression / RL	Quantiles, policy probabilities, online-feedback context and calibration features	Diagnostic contributor
Skill gates	Structured quality, sequence, regime and risk diagnostics	Contributor gates, not execution authority

A* scenario classification

The A* engine remains an internal challenger over CONTINUE, PULLBACK, REVERSAL_ATTEMPT and FAKEOUT transitions. It may inform the council, but Path A/Path B and double-lane public rendering are retired. The operator receives one studied BUY, SELL or MIXED_EVIDENCE regression read with confidence and reasons, while the separate entry answer remains exactly BUY_NOW, SELL_NOW or DO_NOT_ENTER.

Core skill gates

The active source defines thirteen core gates: calibrated probability/conformal interval, discrete FSM state, algorithmic evidence heap, regression error estimation, knowledge representation, candle-group context, formal automata, predictive analytics and related base gates. Its router can learn relative weighting from feedback. The historical comment's consensus recipe (confidence ≥ 0.82 , at least nine gates passing and memory similarity ≥ 0.87) is a diagnostic threshold, not permission to trade.

6. Deep candle intelligence, Pair DNA and historical memory

Every proven completed candle enters a V3 study lane before any directional read is published. Price OHLC is used when available; otherwise normalized-price or pixel-price proxies remain explicitly labelled so chart pixels are never presented as broker prices or pips.

Service	What it stores or measures	Integrity boundary
Candle Intelligence V3	Exact body/range, upper and lower wicks, wick/body ratios, close location, candle type, personality, rejection, sweep, acceptance and relation to the previous candle	Closed candles only; mixed or contradictory coordinate spaces fail closed
Behavioral Sequence V3	UP swing, DOWN swing and REST states; segment candles and seconds; change, efficiency, transitions, major trend and inner trend	Current segment remains open until a later boundary proves completion
Exact candle ledger	SQLite WAL row keyed by pair, timeframe and authoritative closed-candle identity	Upsert prevents duplicate unique candles; WAL + FULL sync + immediate transaction
Pair DNA	Unique chronological candle aggregates, completed swing/rest boundaries, transitions, regimes, objects and non-causal outcome associations	Locks CLOSED_TIMESTAMP_V1 or TRACKER_EVENT_SEQUENCE_V3 per pair/timeframe; conflicting order domains are skipped
Historical similarity	Explainable 60-value sequence fingerprint, same-pair nearest sequences and supported UP/DOWN/REST outcomes	Outcome requires exact resolver N to N+1 plus unique prior-close re-observation on the current axis
Object relationship graph	Explicit candle anchors, observed-with, co-occurrence and proven normalized overlap	No inferred anchor, causal claim, order field or execution authority

The one studied direction

Major regression carries 52% of the directional study weight, inner regression 30%, and a support-qualified historical continuation 18%. Opposing evidence reduces confidence. History stays unavailable until at least three causally labelled similar cases exist, and close probabilities remain MIXED_EVIDENCE rather than being forced into BUY or SELL.

NO AUTOSCALE OR REACQUISITION LEAKAGE

A prior pixel/proxy close is never compared directly with a newly scaled frame. Positional IDs never persist. The current resolver event must be exactly prior sequence +1 and the prior close must be uniquely found again among the current frame's earlier candles; otherwise outcome maturation is skipped.

7. From classification to a trading decision

The decision kernel turns independent evidence into a coherent market read: regime, price location, path clarity, timing, risk, continuation/reversal plausibility and opposing-force room. It produces a candidate directional thesis only. The Book Strategy Master V3 then applies the complete rule set and is the final strategy decider; Model Council V3 contributes freshness, model health, source lock, sequence, timing and promotion evidence around it.



Playbook classification

Decision concept	Examples in the implementation	Effect
Playbook family	SMC turtle soup; BMS/RTO; AMD reversal; supply/demand reaction; trendline/channel; Fibonacci OTE; pivot reaction; slingshot false break; chop/no-trade	Chooses the trading grammar that must fit the observed structure
Reaction grammar	Wick rejection, body acceptance, retest hold, reclaim after sweep, continuation pressure, exhaustion, no reaction	Distinguishes a real response from a merely nearby level
Entry profile	Aggressive sniper, conservative retest, continuation retest, reversal reclaim, momentum acceptance, watch-only, no-trade	Controls the evidence expected before maturity
Maturity state	NO_OPPORTUNITY → EARLY_FORMING → VALID_WATCH → PREPARE → ENTER_NOW; or LATE_CHASE / INVALIDATED / MISSED	Makes timing explicit and prevents late relabeling

Hard blockers and fail-closed behaviors

- Stale, non-advancing or inconsistent live truth; dirty cache; missing required models; unhealthy API; invalid source lock.
- Late chase, invalidated candidate, buy-high / sell-low classification, insufficient path room or opposing force too close.
- Incomplete candle proof should remain WATCH/PREPARE rather than become a false runtime failure; it still cannot be entered.
- The public interface keeps Major trend, Inner trend, Regression study and Entry permission separate so a historical lean cannot be mistaken for a permitted trade.

8. Permission and packet validation: the real execution boundary

A permitted trade is not just BUY or SELL. The current system requires an explicit playbook ENTER_NOW decision, council evidence, a current packet, and a separate schema/runtime validation. The executable packet is built by `execution/packet_v3.py::build_execution_packet_v3` and verified by `validate_execution_packet_v3`.

Required contract group	Validation expectation	Why it exists
Identity	Session, symbol, timeframe, instrument context and source lock are present and match expected values	Prevents wrong-chart / wrong-session action
Freshness	Frame, capture and state versions advance; valid-until is in the future; fresh cache and hashes are supplied	Prevents stale action
Live integrity	is_live, frame_advancing, capture_advancing and state_advancing are true; source is model_council	Separates runtime failure from a market decision
Execution	state EXECUTABLE; side strictly BUY or SELL; expiry matches time sequence; amount action is DO_NOT_CHANGE_AMOUNT	Eliminates legacy ambiguity and local amount control
Council health	final_state/side agree; all required models awake; health provenance is carried	Avoids unsupported/misaligned permission
Allowance	Explicit PG_ALLOWANCE_PACKAGE_V1, accepted and execution-ready	Makes handoff scope explicit

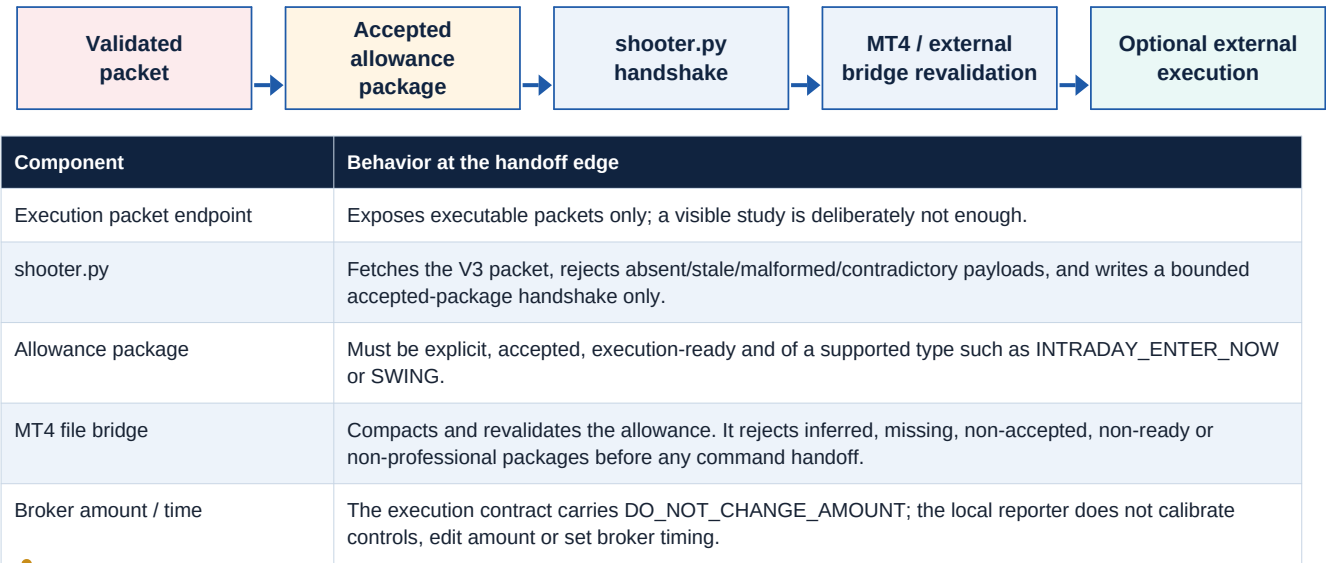
WHAT IS REJECTED

Raw action / execution_action payloads, old schemas, STUDY_PACKETs, CALL/PUT aliases, missing final side, stale packets, non-advancing frames and identity mismatches are not executable packets.

The packet’s authority metadata carries `PLAYBOOK_FINAL_DECIDER_V3` as strategy authority and `PG_EXECUTION_PACKET_V3` as packet authority. Packet validation classifies freshness, identity, cache, live-state and model-awake failures as `RUNTIME_INTEGRITY`—not as a reason to reinterpret unsafe state as a market blocker.

9. What ‘traded’ means in FINAL_LIVE

In the canonical FINAL_LIVE setup, direct local broker clicking is disabled. The launcher may enable live evaluation and construction of permission state, while PHOENIXGUARD_ALLOW_LIVE_BROKER_CLICKS remains disabled. The historical shooter is now a package reporter, not a clicker.



IMPORTANT BOUNDARY

The repository proves a controlled handoff architecture, not a promise of broker fill, price, profit or loss avoidance. Broker rules, spread, slippage, trigger conventions and fill mechanics remain external and variable.

10. Continuous closed-candle research history

PhoenixGuard studies every newly proven pair/timeframe-scoped closed candle exactly once. No operator button creates a study baseline and no preset horizon ends the history. Repeated frames are idempotent; a stopped and restarted capture service resumes only after closed-candle identity and source continuity are proven.

Research capability	What it adds	Hard interpretation boundary
Motif lattice	Single-candle micro-events; 3-5 candle atoms; 7-12 candle compounds; full swing/rest regimes	Bounded historical composition, not a future outcome
Time-to-event	Kaplan-Meier-style next-swing, direction-change and rest-end curves, plus object-conditioned curves from matured Pair DNA	Descriptive duration evidence, not a deadline or cause
Adaptive ontology	Pair-scoped shadow features evaluated on closed-candle train/holdout evidence, with versioned promotion and rollback	Passing means eligible for study only
Exact paths	Anchor-known normalized path, MFE, MAE, efficiency, time in state and a bounded motif-linked trajectory library	Historical reconstruction only
Cross-pair graph	Shared-timestamp normalized Granger-style proxy plus mutual information	Explicitly non-causal association; no influence or entry claim
Concept drift	Bounded distribution test and deterministic regime partition identity	Distribution change, not direction
Proof certificates	Hashes claim, inputs, derivation, closed-candle IDs, coordinate and order domains	Integrity only; not source authentication or execution authority

Continuous integration

MarketStudyServiceV3 publishes bounded motif_lattice, survival_network, path_reconstruction, adaptive_feature_ontology, concept_drift, regime_partition, cross_pair_association and claim_proofs fields from restart-safe history. Pair-scoped ontology audit state persists independently; drift snapshots and append-stable partitions persist inside Pair DNA. The atomic cross-pair coordinator retains only bounded normalized returns; without a genuine synchronized peer or support it reports insufficient evidence and never creates an edge.

Continuity handling

A history row receives a market-study snapshot only when its closed-candle key and monotonic sequence match exactly. Missed visual rollovers enter a fail-closed reacquisition path. Unknown gaps remain explicit; the newest study is never copied backward. Cross-pair evidence additionally requires exact shared contiguous closed timestamps and compatible normalized geometry. Granger-style variance reduction and mutual information always remain non-causal associations.

11. Operator experience, auditability and operational controls

The privacy-safe PG_OPERATOR_WORKSPACE_V1 presents three plain-language answers before technical detail. It removes provider internals, filesystem paths, raw telemetry and private strategy vocabulary from the first viewport. Major/inner trend, regression, continuous history, freshness, exact-frame media, stream health and overlay geometry remain supporting evidence in progressively disclosed detail.

Primary answer	Public contract
Market origin and history	Explain the completed movement, major trend, swing/rest rhythm and the closest supported historical behavior.
Studied direction then and now	Separate the prior studied side from the current major/inner read, maturity, confirmation and invalidation.
Best decision right now	Publish the best current action: BUY_NOW, SELL_NOW, watch a named pullback/rally, hold/protect, avoid chasing, or wait—with the decisive reason and change condition.

Audit and certification

- Runtime artifacts: session.json, compact_live_state.json, display_state.json, events.jsonl, continuous study stores, exact frame media and overlays.
- Evidence loop: allowed-entry and blocked-ENTER_NOW screenshots, future outcome scoring, progression galleries, manifests and forensic reports.
- Health and topology: API health, single runtime/venv validation, model warm state, source lock, dashboard hydration, broker freshness and runtime traces.
- Useful live endpoints: /v1/mobile/health; operator/state/v1/{session}; model-council execution/latest; runtime/trace/v3; and the /v3 dashboard.

OPERATING RULE

An absent or invalid execution packet makes the entry answer DO_NOT_ENTER and names the failing gate. It does not erase the studied direction, historical explanation or supported setup with an unexplained WAIT.

12. Implementation map and verification checklist

The following files are the most relevant active surfaces for understanding or changing the system. This blueprint intentionally prioritizes current source and canonical blueprints over archived architecture snapshots.

Area	Primary active source / canonical reference
Launch / topology	Backend/launch/launch_phoenixguard_live_ready.ps1; start_phoenixguard_full_local.ps1; start_phoenixguard_24_7_tracker.py
Live tracker	Backend/src/phoenixguard/mobile_api/window_tracker.py
CPU stream observer	Backend/src/phoenixguard/vision/cpu_stream_v3.py; window_tracker.py producer/consumer integration
Closed-candle resolver	phoenixguard/decision/scene_forecast_contributor_v3.py; mobile_api/window_tracker.py resolver-to-study bridge
Deep V3 study	phoenixguard/study/candle_intelligence_v3.py; behavioral_sequence_v3.py; candle_ledger_v3.py; pair_dna_v3.py; historical_similarity_v3.py; object_relationship_graph_v3.py
Advanced V3 research	phoenixguard/study/motif_lattice_v3.py; adaptive_feature_ontology_v3.py; concept_drift_v3.py; cross_pair_association_v3.py; cross_pair_coordinator_v3.py; study_claim_proof_v3.py
API / public state	Backend/src/phoenixguard/mobile_api/app.py; live_state_v3.py
Inference	Frontend/dashboard/main.py::run_inference
Vision / overlays	phoenixguard/vision/*; tracking/market_object_tracker_v3.py; vision/v3_overlay_contract.py
Decision	phoenixguard/decision/model_council_v3.py; book_strategy/*; scenario_decision_kernel.py; a_star_scenarios.py; skill_gates.py
Execution boundary	phoenixguard/execution/packet_v3.py; v3_language.py; Backend/launch/shooter.py; phoenixguard_mt4_file_bridge.py
Dashboard	Frontend/dashboard/static/window_tracker_dashboard.html
Canonical specs	docs/architecture/PHOENIXGUARD_V3_MARKET_STUDY_BLUEPRINT.md; PhoenixGuard_System_Blueprint.md; docs/active_execution_paths.md; docs/execution_packet_schema_matrix.md

Before considering a live stack healthy

- Health returns 200 and exactly one logical stack owns the configured port.
- The session is running; stream/frame/capture/state identities advance; pair, timeframe, HWND and geometry generation remain consistent.
- The CPU observer sustains its declared sample rate without unbounded memory, pending backlog or overlapping heavy studies.
- Latest window/chart/overlay artifacts, operator state and dashboard all return successfully and agree on version identity.
- Continuous study requires no manual baseline; only genuinely new identity-proven closed candles advance it.
- Model health, source lock, packet freshness and package handoff status are checked separately from market analysis.
- No regression read, overlay or raw signal is treated as entry permission; only a fresh validated V3 packet plus accepted allowance may cross the handoff boundary.

BOTTOM LINE

PhoenixGuard is an evidence-to-permission system, not a raw-frame-to-click system. Observe continuously, study material events, classify explicitly, state the best useful action honestly, and permit handoff only through independently validated current truth.